

Railway Crew Management System (RCMS)

Professional Product Requirements Document (PRD) v1.0

Prepared for Backend Development

1. Executive Summary

The Railway Crew Management System (RCMS) is a secure web application for managing railway crew operations from CMS import through sign-on, operational updates, archival, and auditing. The system is designed with a layered architecture using FastAPI, SQLAlchemy, PostgreSQL, Alembic, and JWT authentication.

2. Objectives

- Digitize crew management workflow.
- Reduce manual data handling.
- Provide secure role-based access.
- Maintain complete audit trails.
- Provide scalable REST APIs.

3. Scope

Included: Authentication, Users, Crew, CMS Import, Sign-On, Main Data, Archive, Audit Logs, Import Logs, RBAC, Swagger APIs.

Excluded (v1): AI modules, notifications, analytics, train/station master, shift/assignment/availability unless requested by client.

4. Technology Stack

Layer	Technology
Backend	FastAPI
ORM	SQLAlchemy 2.x
Database	PostgreSQL
Migration	Alembic
Authentication	JWT
Testing	Pytest
Docs	Swagger/OpenAPI

5. User Roles

Role	Responsibilities
Admin	Full system administration
Controller	Operational updates
Supervisor	Monitoring and reporting
Lobby Operator	Crew sign-on operations
Viewer	Read-only access

6. Business Workflow

CMS Import → cms_data → Crew Sign-On → signon_data → main_data → Controller Updates → archive_data → audit_logs

7. Core Database Tables

- users

- crew
- cms_data
- signon_data
- main_data
- archive_data
- audit_logs
- import_logs

8. Backend Architecture

API Layer → Service Layer → Repository Layer → SQLAlchemy Models → PostgreSQL

9. Functional Requirements

- JWT Authentication
- Role-Based Access Control
- Crew CRUD
- CMS Import
- Sign-On Validation
- Archive Workflow
- Audit Logging
- Pagination & Filtering
- Swagger Documentation
- Alembic Migrations

10. Non-Functional Requirements

- Security
- Scalability
- Maintainability
- Performance
- Reliability
- Transaction Safety

11. Acceptance Criteria

- Complete Crew → CMS → Sign-On → MainData → Archive workflow
- All APIs authenticated
- Audit logs generated
- Integration tests passing
- Database migrations stable

12. Future Enhancements

- Frontend Dashboard
- Reporting Module
- Notifications

- Analytics
- Mobile Support (if requested)

13. Conclusion

Version 1.0 focuses on delivering a clean, maintainable, production-ready backend aligned with the approved business workflow and avoiding speculative features not requested by the client.